Do Tuan Minh

tuanminhdo1203@gmail.com | +84 834 134 509 | github.com/TuanMinhDo1203

Final-year Artificial Intelligence Undergraduate | Computer Vision, Document AI, Edge AI, and Applied AI Systems

Profile

Final-year Artificial Intelligence undergraduate (GPA approximately 8.8/10) working across computer vision, document AI, edge deployment, retrieval, and applied AI systems. First author of multiple peer-reviewed conference papers, including an APWeb-WAIM 2026 paper receiving the 3rd Best Research Paper Award; co-author of a 2026 Q1 journal article. Current research studies detector-aware low-light enhancement. Industry and team projects span OCR workflows, model optimization, and AI-assisted enterprise applications.

Education

FPT University, B.S. in Artificial Intelligence — Sept 2023 – Early 2027 expected

- GPA: 8.8/10
- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Text Mining, Data Structures and Algorithms
- Research Interests: Computer Vision, Low-Light Image Enhancement, Object Detection, Document AI/OCR, Efficient Vision and Edge AI

Industry Experience

FPT Information System (FIS), AI Engineer Intern — Oct 2025 – Apr 2026

- Developed OCR post-processing modules to reconstruct structured **.docx** and **.html** documents from OCR JSON outputs while preserving layout consistency.
- Curated and preprocessed domain-specific Vietnamese document datasets for VietOCR fine-tuning, evaluation, and error analysis.
- Supported distributed OCR inference workflows using **Redis, RabbitMQ, Docker**, and GPU workers on internal NVIDIA H100 infrastructure.
- Built centralized logging and monitoring components using **Docker Compose, Elasticsearch, and Grafana** to track inference latency and system performance.
- Worked on table-layout detection and asynchronous document-processing workflows with senior engineers; contributed to components rather than overall platform architecture.

FPT Software, Enterprise Application / AI Engineering Intern — September 2026 – Present

- Worked with **OutSystems Developer Cloud (ODC)** on enterprise application development, business workflows, data modeling, and application logic.
- Explored **Azure AI Document Intelligence** (Custom Extraction Models), including document data labeling, custom model training, and testing for structured field extraction.
- Gained hands-on exposure to API integration and AI-assisted enterprise workflows, including chatbot and automation-related initiatives.

Selected Publications

- **First Author** — *Recognition of Handwritten Python Code in Academic Exam Settings Using a Multi-stage OCR Pipeline*. APWeb-WAIM 2026 (Research Track), Springer LNCS, Vol. 16995. [DOI]. **3rd Best Research Paper Award.** Developed a multi-stage handwritten code OCR pipeline with indentation reconstruction, achieving **90.51% line recovery** and **27.01% exact match**.
- **First Author** — *CPGA-Stable: Improving Optimization Stability in Physics-Inspired Low-Light Image Enhancement*. MIWAI 2026, Springer LNAI, Vol. 17147. [DOI].
  Improved optimization stability through bounded parameterization and initialization, increasing mean PSNR from **14.75 to 20.09 dB** while reducing run-to-run variance.
- **Co-author** — *Towards Accurate and Efficient Waste Image Classification: A Hybrid Deep Learning and Machine Learning Approach*.
  Ain Shams Engineering Journal, Elsevier, **SCI Q1**, 2026. [DOI].

Investigated hybrid feature learning and classification, achieving **over 95% feature reduction** with up to **99.72–100% accuracy** across evaluated datasets.

- **Co-author** — *Impact of DICOM Preprocessing Strategies on Chest X-ray Abnormality Detection*.
  APWeb-WAIM 2026 (Research Track), Springer LNCS, Vol. 16995. [DOI].
  Benchmarked **six DICOM preprocessing strategies** across YOLOv8, Faster R-CNN, and RT-DETR, demonstrating detector-dependent effects on abnormality detection.
- **First Author** — *Understanding Loss Function Behavior Across Lightweight Low-Light Image Enhancement Architectures*.
  EIDT-SIMC 2026. **Accepted.**
  Investigated how different loss functions interact with lightweight enhancement architectures through controlled comparative experiments.

Additional Publications

- **First Author** — *Robustness and Vietnamese-Specific OCR Error Analysis on Receipt Images*.
  MISNC 2026, Springer CCIS, Vol. 3091. [DOI].
- **First Author** — *Comparative Study of Deep Learning Architectures for Water Level Forecasting in the Red River*.
  MEDES 2025, Springer CCIS, Vol. 3103. [DOI]. **Best Paper Award.**
- **First Author** — *Comparative Analysis of Loss Functions in Deep Learning Models for Water Level Forecasting*.
  ICAIDES 2025, IEEE. [DOI].

Selected Projects

EXAM_OCR – Handwritten Code OCR Platform Jan 2026 – Present

- Developed an end-to-end platform for handwritten Python code recognition, combining line detection, line classification, TrOCR recognition, and indentation-aware reconstruction.
- Conducted component-level error analysis, identified recognition as the main end-to-end bottleneck, and evaluated trade-offs among accuracy, latency, and deployment constraints.
- Reduced CPU inference latency from **81.7s to 23.4s** (approximately 71.4%) through PyTorch-to-OpenVINO optimization; measured mean CER approximately 0.2405 and exact match 27.01%.
- Built asynchronous processing and teacher-review workflows using **FastAPI, Celery, Redis, Azure SQL, Azure Blob Storage, and Docker**.
- GitHub: `github.com/TuanMinhDo1203/EXAM_OCR`

EdgePlant – Agricultural Disease Detection on Edge Hardware August 2026

- Contributed to deployment of a 116-class YOLO11x model on Qualcomm QCS8550, converting PyTorch to ONNX and Qualcomm QNN with W8A16 quantization and hardware acceleration.
- Reduced the model artifact from over 400 MB to an approximately 58.8 MB checkpoint.

StallyApp – AI-Assisted B2B Food and Beverage Procurement April 2026 - July 2026

- Co-developed a human-reviewed procurement workflow from purchase request and supplier discovery through RFQ, quotation comparison, negotiation drafts, purchase-order approval, and inventory receiving.
- Integrated AI assistance for purchase-request drafting, supplier reranking, and multimodal quotation extraction using Gemini and Google Search grounding; worked with React, TypeScript, Express, SSE, and Supabase/Postgres.

Student Academic Assistant Chatbot – Intent-Aware RAG May 2025 – Aug 2025

- Developed an intent-aware RAG system for academic support using **BGE-M3 embeddings, Qdrant**, metadata filtering, and structured academic data.
- Implemented a DistilBERT intent classifier for pre-retrieval filtering and designed an evaluation protocol using Precision, Recall, F1, and latency.
- Compared local and API-based intent classification: DistilBERT achieved **F1 = 0.60** at 0.9s/query, while Gemini achieved **F1 = 0.83** at 2.5s/query.
- GitHub: `github.com/TuanMinhDo1203/FPT_FAP_CHAT`

Reddit Sentiment Analysis – Premier League 2024/2025 Jan 2025 – Mar 2025

- Analyzed Reddit discussions on Premier League matches using a fine-tuned BERT-based sentiment classification model.
- Collected Reddit and match-related data through web crawling and visualized sentiment trends using a Streamlit dashboard.

- Tools: Python, BeautifulSoup, Selenium, Streamlit, Transformers.
- GitHub: `github.com/TuanMinhDo1203/Reddit_Sentiment_Analysis`

Honors and Awards

- **APWeb-WAIM 2026:** 3rd Best Research Paper Award for *Recognition of Handwritten Python Code in Academic Exam Settings Using a Multi-stage OCR Pipeline* (first author).
- **MEDES 2025:** Best Paper for *Comparative Study of Deep Learning Architectures for Water Level Forecasting in the Red River* (first author).
- **OlpAI 2025 – Vietnam Student Artificial Intelligence Olympiad:** Second Prize, Central Vietnam Regional Round; National Finalist with Consolation Prize.
- **SoICT Summer School on Edge AI 2026:** 1st Runner-up Presentation Award for *Smart Agricultural Disease Diagnosis System with AI Integration*, Hanoi University of Science and Technology.
- **FPT Research Connect (ResConnect) 2026: Second Prize (Creativity Award), Information Technology Track** – among 41 finalist teams across four academic disciplines.
- **Top 100 Outstanding Students**, FPT University: Fall 2024, Summer 2025, Fall 2025, Spring 2026.

Technical Skills

Programming / ML: Python, SQL, PyTorch, TensorFlow, scikit-learn, Transformers
Vision / Document AI: OpenCV, YOLO, ResNet, TrOCR, VietOCR, DICOM preprocessing
Retrieval / NLP: RAG, Qdrant, BGE-M3, Gemini API, DistilBERT
Backend / Infrastructure: FastAPI, Celery, Redis, RabbitMQ, Docker, Elasticsearch, Grafana
Deployment: OpenVINO, ONNX Runtime, CUDA, Qualcomm QNN
Research: Experimental design, model benchmarking, ablations, error analysis

Languages

Vietnamese: Native | **English:** TOEFL iBT 82/120 (2023)